

3.5 WHY JOINT TRAINING ACQUIRES SCALE SLOWLY

Can joint training acquire the scales needed for high precision? After five million updates on mixed sine, joint Adam reaches relative error 1.81×10^{-3} , versus 6.62×10^{-7} on supplied features. Its RMS relative slope reaches 0.0582, compared with the supplied 1/4; GD acquires less scale (Figure 4). The preceding accuracy results motivate explaining this slow population movement.

The surviving force. For squared loss $L = \frac{1}{2}\|q_\theta - f\|^2$, splitting output into its affine part and orthogonal remainder gives $\nabla L = R + F$. The *tracking gradient* R measures departure from coarse equilibrium. The *effective fine gradient* F combines the non-affine residual gradient with the compensation needed to preserve coarse output. In the audited GD regime, tracking is small; compensation remains within F . We study the ensuing effective flow $\dot{\theta} = -F$ and account for tracking and finite GD steps through explicit disturbance allowances.

Why weak force persists. Let e_H be the non-affine residual, $J_H = D_\theta e_H$, and $v = F/\|F\|$. Along effective flow,

$$\frac{d}{dt} \log \|F\| = - \underbrace{\|J_H v\|^2}_{\text{residual relaxation}} + \underbrace{\kappa_H + \kappa_C}_{\text{geometry and compensation feedback}}. \quad (1)$$

Relaxation fits away the error driving the force; feedback changes sensitivity to the remaining error. Small F means weak residual coupling after compensation. Rapid acquisition requires this coupling to strengthen. On smooth step, force grows 2.19-fold while error remains near 49% (Figure 5): reinforcement occurs, but remains limited. We bound its accumulation by B_t , using the population’s second output response weighted by residual and compensation norms.

Theorem 3.3 (Slow scale acquisition, informal). *Restart GD after tracking becomes small. Let F_0 and E_0 be the restart effective gradient and non-affine residual norm, and $t = n\eta$. If accumulated feedback is bounded by B_t , then, under the conditions in Appendix A,*

$$\|q_{\theta_n} - f\|^2 \geq E_0^2 - 2te^{2B_t}\|F_0\|^2 - \Delta_{\text{err}}(t), \quad (2)$$

$$\lambda_{\text{RMS}}(t) \leq \lambda_{\text{RMS}}(0) + \frac{h}{\sqrt{W}}te^{B_t}\|F_0\| + \Delta_{\text{scale}}(t). \quad (3)$$

Here $\lambda_{\text{RMS}} = h\|\gamma\|_2/\sqrt{W}$; h is the reference spacing. The allowances cover tracking and finite steps and vanish for effective flow.

The proof bounds force amplification, then integrates force to bound population travel and its square to bound output improvement. A complementary theorem closes the mechanism: if population travel produces limited additional reinforcement and accumulated force concentration is controlled, weak force supplies too little movement to amplify itself rapidly. Its finite-time condition and first-exit proof are in Appendix A.4; parameters remain free to evolve.

Empirical check. The broader feedback condition covers all 46 sampled paths across 23 targets over 20k further updates. Six dense continuations over 100k further updates give RMS growth bounds below 8×10^{-4} and relative-error floors above 39%, with close GD/effective-flow agreement. The stronger loop criterion passes all six shorter continuations and three longer ones (Appendix B). It explains a sufficient mechanism for persistence; the broader theorem retains longer coverage.

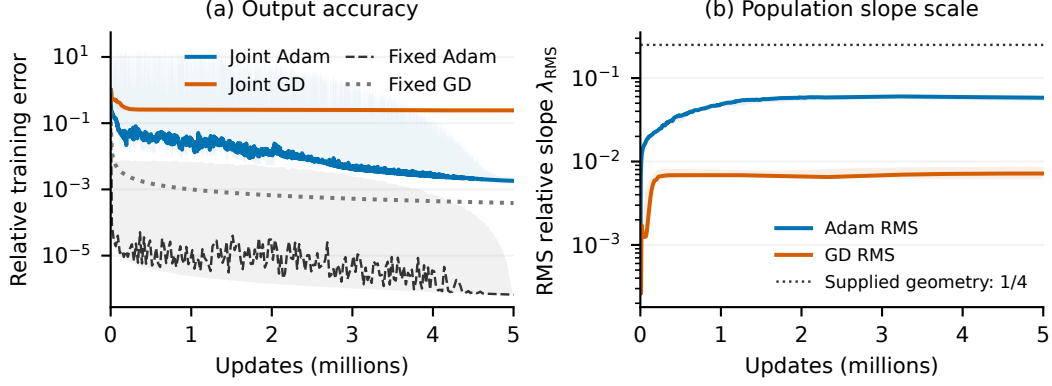


Figure 4: **Limited scale acquisition accompanies the precision gap.** Mixed sine, width 512, five paired seeds, five million full-batch updates. Left: relative training errors with trained readouts; fixed Adam/GD use uniform $\lambda = 1/4$ features. Right: RMS relative slopes, $h = 2/467$. Bands retain seed variation and within-bin extrema. Joint and fixed-Adam recipes use final-validation selection (Appendix B).

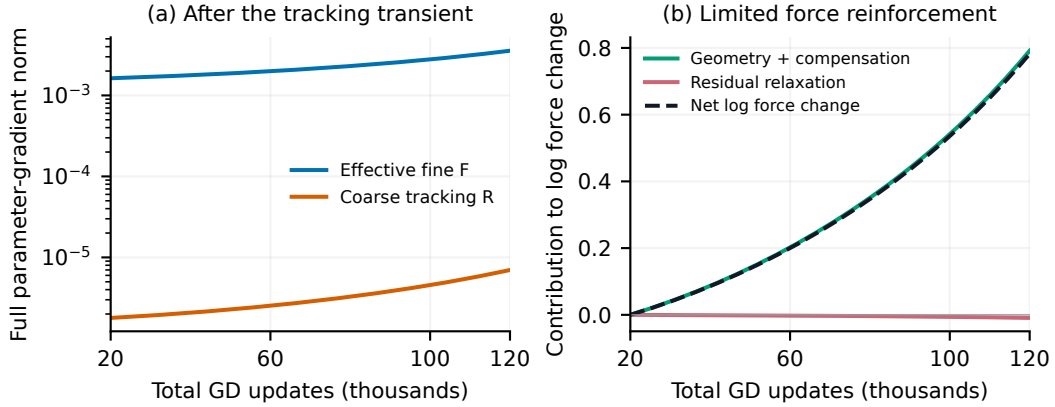


Figure 5: **Weak effective force survives positive reinforcement.** Matched smooth-step continuations, width 705, from age 20k to 120k. Left: GD full-parameter gradient norms; tracking stays below 0.2% of effective force at saved times. Right: cumulative terms in (1) along paired effective flow. Both axes use total GD updates with $\eta = 0.002$; paired force norms differ by less than 0.08%.

A Conditional persistence under evolving geometry

This appendix makes Theorem 3.3 precise. The only structural feedback assumption is an aggregate integral along the evolving path. No neuronwise speed bound, fixed Jacobian, or stationary slope equilibrium is required. The conclusion concerns a specified interval; entry into the regime from random initialization is a separate question.

A.1 OUTPUT GEOMETRY AND THE EXACT FORCE DECOMPOSITION

Let μ be the equally weighted empirical measure on training inputs in $[-1, 1]$, with nonzero variance. Output inner products use $L^2(\mu)$; parameter space has its Euclidean metric. For $f \in L^2(\mu)$, set

$$q_\theta(x) = b + \sum_{j=1}^W w_j \tanh(\gamma_j x + \beta_j), \quad L(\theta) = \frac{1}{2} \|q_\theta - f\|^2. \quad (\text{A.1})$$

All parameter blocks evolve. Let Q_C embed two orthonormal affine coordinates into output space, and $P_H = I - Q_C Q_C^*$. Define

$$e_C = Q_C^*(q_\theta - f), \quad e_H = P_H(q_\theta - f), \quad J_C = D_\theta e_C, \quad J_H = D_\theta e_H. \quad (\text{A.2})$$

Thus “fine” means the entire orthogonal complement of affine output, independently of the target. Writing $u_j = \gamma_j x + \beta_j$, the full output Jacobian $J = D_\theta q_\theta$ has columns

$$J_{\gamma_j} = w_j x \operatorname{sech}^2 u_j, \quad J_{\beta_j} = w_j \operatorname{sech}^2 u_j, \quad J_{w_j} = \tanh u_j, \quad J_b = 1. \quad (\text{A.3})$$

In particular $J_C = Q_C^* J$ and $J_H = P_H J$ include the evolving geometry and readout sensitivities.

Suppose J_C has full row rank along the interval under study. Define

$$\begin{aligned} \Pi &= I - J_C^*(J_C J_C^*)^{-1} J_C, & \ell &= (J_C J_C^*)^{-1} J_C J_H^* e_H, \\ F &= \Pi J_H^* e_H = J_H^* e_H - J_C^* \ell, & R &= J_C^*(e_C + \ell). \end{aligned} \quad (\text{A.4})$$

Then $\nabla L = F + R$ exactly. The compensating gradient $-J_C^* \ell$ is part of F ; it does not disappear with tracking. Since $J_C F = 0$, effective flow $\dot{\theta} = -F$ preserves the coarse output already present at the restart, while all parameter blocks continue to move.

A.2 A STRUCTURAL MEASURE OF REINFORCEMENT

For $F \neq 0$, put $v = F/\|F\|$ and define

$$\kappa_H = -\langle e_H, D^2 e_H[v, v] \rangle, \quad \kappa_C = \ell \cdot D^2 e_C[v, v], \quad (\text{A.5})$$

$$d_{\text{dir}} = \|e_H\| \|D^2 e_H[v, v]\| + \|\ell\| \|D^2 e_C[v, v]\|. \quad (\text{A.6})$$

Consequently $\kappa_H + \kappa_C \leq d_{\text{dir}}$, without a favorable sign assumption. This rate uses the network’s second output response in a unit parameter direction. It is not computed by dividing observed net force growth by the force. Let B_t be a finite, nondecreasing allowance with $B_0 = 0$ satisfying

$$\int_0^t d_{\text{dir}}(s) ds \leq B_t \quad \text{for every } t \in [0, T]. \quad (\text{A.7})$$

At zeros of F , set $d_{\text{dir}} = 0$; the effective-flow contribution to its norm derivative is zero there.

Proposition A.1 (Force and residual identities). *Along effective flow, wherever $F \neq 0$,*

$$\frac{d}{dt} \log \|F\| = -\|J_H v\|^2 + \kappa_H + \kappa_C, \quad \frac{d}{dt} \|e_H\|^2 = -2\|F\|^2. \quad (\text{A.8})$$

Proof. Write $g_H = J_H^* e_H$. Differentiating $F = \Pi g_H$ along $\dot{\theta} = -F$ gives

$$\frac{1}{2} \frac{d}{dt} \|F\|^2 = -F^* D\Pi[F] g_H - \langle e_H, D^2 e_H[F, F] \rangle - \|J_H F\|^2. \quad (\text{A.9})$$

The projector identity implies $\Pi D\Pi[F] \Pi = 0$. Differentiating $\Pi J_C^* = 0$ and using $g_H = F + J_C^* \ell$ yields $F^* D\Pi[F] g_H = -\ell \cdot D^2 e_C[F, F]$. Dividing by $\|F\|^2$ proves the first identity. For the second, $\frac{d}{dt} \|e_H\|^2 = -2 \langle J_H^* e_H, F \rangle = -2 \|F\|^2$, since Π is an orthogonal projector. $\square$

Both identities hold for the moving nonlinear network. The negative term fits away existing residual; the signed curvature terms describe changes in sensitivity and coarse compensation. Positive feedback is compatible with slow acquisition when its accumulation permits only modest amplification of a weak starting force.

A.3 POPULATION ACQUISITION AND OUTPUT ERROR, INCLUDING DISTURBANCES

Theorem A.1 (Conditional aggregate persistence). *Let $\theta : [0, T] \rightarrow \mathbb{R}^{3W+1}$ be absolutely continuous and satisfy $\dot{\theta} = -F(\theta) - S$ almost everywhere. Assume the coarse projector is defined along the path and (A.7) holds. Set $s_0 = \|F(\theta(0))\| > 0$, $E_0 = \|e_H(\theta(0))\|$, and*

$$\begin{aligned} U(t) &= \int_0^t \|DF[S]\| ds, & V(t) &= \int_0^t \|S\| ds, & Z(t) &= \int_0^t [\langle e_H, J_H S \rangle]_+ ds, \\ A(t) &= te^{B_t} [s_0 + U(t)] + V(t). \end{aligned} \quad (\text{A.10})$$

Assume these integrals are finite; nondecreasing upper budgets may replace them. For a reference spacing $h > 0$, let $\lambda_j = h|\gamma_j|$ and

$$p_\delta(t) = \frac{1}{W} \# \left\{ j : \sup_{0 \leq s \leq t} (\lambda_j(s) - \lambda_j(0)) \geq \delta \right\}, \quad \delta > 0. \quad (\text{A.11})$$

Then, for every $t \leq T$,

$$\|F(\theta(t))\| \leq e^{B_t} [s_0 + U(t)], \quad (\text{A.12})$$

$$\int_0^t \|\dot{\theta}(s)\| ds \leq A(t), \quad (\text{A.13})$$

$$\|q_{\theta(t)} - f\|^2 \geq [E_0^2 - 2te^{2B_t} [s_0 + U(t)]^2 - 2Z(t)]_+, \quad (\text{A.14})$$

$$\lambda_{\text{RMS}}(t) \leq \lambda_{\text{RMS}}(0) + \frac{hA(t)}{\sqrt{W}}, \quad (\text{A.15})$$

$$p_\delta(t) \leq \min \left\{ 1, \frac{h^2 A(t)^2}{W\delta^2} \right\}. \quad (\text{A.16})$$

Proof. The force identity gives, almost everywhere along the disturbed path,

$$\frac{d}{dt} \|F\| \leq d_{\text{dir}} \|F\| + \|DF[S]\|. \quad (\text{A.17})$$

The inequality extends across zeros of F in the absolutely continuous norm inequality: the effective contribution $DF[F]$ vanishes there, and the remaining contribution has norm at most $\|DF[S]\|$. Writing $\mathcal{B}(t) = \int_0^t d_{\text{dir}}(s) ds$, the integrating-factor inequality gives

$$\|F(t)\| \leq e^{\mathcal{B}(t)} \left[s_0 + \int_0^t e^{-\mathcal{B}(s)} \|DF[S]\| ds \right] \leq e^{B_t} [s_0 + U(t)]. \quad (\text{A.18})$$

Monotonicity of B, U and the triangle inequality give $\int_0^t \|\dot{\theta}\| \leq te^{Bt}[s_0 + U(t)] + V(t) = A(t)$. The exact disturbed residual identity is

$$\frac{d}{dt} \|e_H\|^2 = -2\|F\|^2 - 2\langle e_H, J_H S \rangle. \quad (\text{A.19})$$

Integrating, bounding the positive error-reducing disturbance by Z , and using $\|q_\theta - f\| \geq \|e_H\|$ proves (A.14).

For population acquisition, put $a_j(t) = \int_0^t |\dot{\gamma}_j(s)| ds$. Minkowski's inequality gives

$$\left(\sum_j a_j(t)^2 \right)^{1/2} \leq \int_0^t \|\dot{\gamma}(s)\|_2 ds \leq A(t). \quad (\text{A.20})$$

The triangle inequality gives $\|\gamma(t)\|_2 \leq \|\gamma(0)\|_2 + A(t)$, proving the RMS bound. Each label counted by p_δ needs $a_j \geq \delta/h$; counting these labels proves (A.16). $\square$

The main theorem follows with explicit allowances

$$\begin{aligned} \Delta_{\text{err}}(t) &= 2te^{2Bt}(2s_0U(t) + U(t)^2) + 2Z(t), \\ \Delta_{\text{scale}}(t) &= \frac{h}{\sqrt{W}}[te^{Bt}U(t) + V(t)]. \end{aligned} \quad (\text{A.21})$$

For $f \neq 0$, relative error stays above ε throughout $[0, T]$ if

$$2Te^{2Bt}[s_0 + U(T)]^2 + 2Z(T) < E_0^2 - \varepsilon^2\|f\|^2. \quad (\text{A.22})$$

If fraction p_0 starts above $\lambda_0 < \lambda_*$, the fraction ever reaching λ_* is at most $\min\{1, p_0 + p_{\lambda_* - \lambda_0}(t)\}$. These conclusions need no maximum-neuron or force-concentration premise. Centers can move freely: the proof uses regular (γ, β, w) coordinates, without division by a slope.

Exact inclusion of discrete GD. For $g = \nabla L$ and $\theta_{n+1} = \theta_n - \eta g(\theta_n)$, use the linear interpolant $\theta(t) = \theta_n - (t - n\eta)g(\theta_n)$. On each step, it satisfies $\dot{\theta} = -F(\theta) - S$ with

$$S(t) = R(\theta(t)) + g(\theta_n) - g(\theta(t)). \quad (\text{A.23})$$

The two terms are tracking and the finite-step defect. The theorem applies at every iterate when their budgets and the feedback condition hold along the interpolant. Effective flow is therefore a special case of the conditional theorem, not an identification of GD with a continuous ODE. The numerical effective-flow bounds below set $S = 0$ and compare GD separately; they are not certified evaluations of all GD allowances.

RMS level versus displacement. The same travel bound gives

$$\left(\frac{1}{W} \sum_j |\lambda_j(t) - \lambda_j(0)|^2 \right)^{1/2} \leq \frac{hA(t)}{\sqrt{W}}. \quad (\text{A.24})$$

This endpoint displacement, shown in Figure S3, differs from the RMS level in Figure 4. The change in RMS level is at most the RMS displacement by the reverse triangle inequality. Integrating the time-dependent force envelope instead of its terminal bound can sharpen the estimates without changing the structural assumption.

A.4 HOW LIMITED POPULATION TRAVEL SUSTAINS WEAK REINFORCEMENT

The preceding theorem turns a feedback budget into acquisition and error bounds. This companion result supplies a sufficient mechanism for that budget. Its assumption measures how much additional reinforcement population movement can generate. The proof couples movement and reinforcement without freezing either.

Write $s(t) = \|F(\theta(t))\|$ and $p_j = (\gamma_j, \beta_j, w_j)$. Let F_j denote the corresponding hidden-parameter block of F , and set

$$I_F = \frac{W \sum_j \|F_j\|^4}{s^4}, \quad \mathcal{C}(t) = \int_0^t \sqrt{I_F(u)} du, \quad \mathcal{A}_4(t) = \int_0^t I_F(u)^{1/4} s(u) du. \quad (\text{A.25})$$

Set $I_F = 0$ at zero force. The denominator includes output-bias force. The quantity $\mathcal{A}_4$ counts accumulated population travel in a fourth-power norm, including reversals; it is not endpoint displacement. In particular it bounds the increase of $M_4^{1/4}$, where $M_4 = W \sum_j \|p_j\|^4$.

Let $\mathcal{B}(t) = \int_0^t d_{\text{dir}}(u) du$. Assume, for constants $b_0, K \geq 0$ and a finite nondecreasing allowance C with $C(0) = 0$,

$$\mathcal{B}(t) \leq b_0 t + K \int_0^t \mathcal{A}_4(u) du, \quad \mathcal{C}(t) \leq C(t), \quad 0 \leq t \leq T. \quad (\text{A.26})$$

The baseline b_0 allows reinforcement already present at the restart; K limits the additional reinforcement generated by population travel. Both assumptions are accumulated conditions. Small force alone does not imply a useful K : changing force direction can expose different output curvature. The empirical audit below tests this response relation.

Theorem A.2 (Persistence through limited feedback from population travel). *Consider effective flow on $[0, T]$ with a defined coarse projector, $s_0 = \|F(0)\| > 0$, and (A.26). Define*

$$H_0(t) = \int_0^t e^{2b_0 u} du, \quad G(T) = \int_0^T \sqrt{C(u)H_0(u)} du. \quad (\text{A.27})$$

If $K s_0 G(T) < r e^{-r}$ for some $r > 0$, then, throughout $[0, T]$,

$$\mathcal{B}(t) < b_0 t + r, \quad s(t) \leq e^r s_0 e^{b_0 t}, \quad (\text{A.28})$$

$$\mathcal{A}_4(t) \leq e^r s_0 \sqrt{C(t)H_0(t)}, \quad M_4(t)^{1/4} \leq M_4(0)^{1/4} + e^r s_0 \sqrt{C(t)H_0(t)}, \quad (\text{A.29})$$

$$\|q_{\theta(t)} - f\|^2 \geq [E_0^2 - 2e^{2r} s_0^2 H_0(t)]_+. \quad (\text{A.30})$$

Moreover, $\lambda_{\text{RMS}}(t) - \lambda_{\text{RMS}}(0) \leq h e^r s_0 \int_0^t e^{b_0 u} du / \sqrt{W}$.

Proof. Suppose $\mathcal{B}(t) - b_0 t$ first reaches r at $\tau \leq T$. The force identity implies $s(u) \leq e^r s_0 e^{b_0 u}$ up to that time. For $t \leq \tau$, Cauchy–Schwarz gives

$$\mathcal{A}_4(t) \leq \sqrt{\mathcal{C}(t) \int_0^t s(u)^2 du} \leq e^r s_0 \sqrt{C(t)H_0(t)}. \quad (\text{A.31})$$

The response premise then gives the contradiction

$$r = \mathcal{B}(\tau) - b_0 \tau \leq K \int_0^\tau \mathcal{A}_4(u) du \leq K e^r s_0 G(T) < r. \quad (\text{A.32})$$

Thus the force and travel bounds hold throughout the interval. Minkowski’s inequality proves the moment bound. Integrating $\frac{d}{dt} \|e_H\|^2 = -2s^2$ proves the error bound; integrating s bounds Euclidean travel and hence RMS slope growth as in Theorem A.1. At a zero of F , effective flow remains stationary by uniqueness, and the conclusions continue to hold. $\square$

For $C(t) = ct$ and $r = 1$, a simpler sufficient condition is

$$\frac{e}{2} K s_0 \sqrt{c} e^{b_0 T} T^2 < 1. \quad (\text{A.33})$$

Indeed $H_0(t) \leq t e^{2b_0 t}$, so $G(T) \leq \sqrt{c} e^{b_0 T} T^2 / 2$. With $b_0 T$ bounded and the other structural constants fixed, this permits durations proportional to $(K s_0 \sqrt{c})^{-1/2}$. Weak initial coupling delays the reinforcement loop; the population need not contract or approach equilibrium.

Disturbances and GD. On $\dot{\theta} = -F - S$, keep U, V, Z from (A.10) and define

$$V_4(t) = \int_0^t \left(W \sum_j \|S_j(u)\|^4 \right)^{1/4} du. \quad (\text{A.34})$$

Replace $\mathcal{A}_4$ in the response premise by $\mathcal{A}_4 + V_4$, and put $s_* = s_0 + U(T)$. If

$$K \left[e^r s_* G(T) + \int_0^T V_4(u) du \right] < r, \quad (\text{A.35})$$

the same first-exit proof gives $\mathcal{B}(t) < b_0 t + r$ and $s(t) \leq e^r s_* e^{b_0 t}$. The error floor becomes $[E_0^2 - 2e^{2r} s_*^2 H_0(t) - 2Z(t)]_+$; Euclidean travel is at most $e^r s_* \int_0^t e^{b_0 u} du + V(t)$. To see this, the integrating-factor bracket is bounded by s_* up to the proposed exit; Cauchy–Schwarz bounds $\mathcal{A}_4$, and V_4 adds disturbed population travel. Substitution in the response premise contradicts (A.35). For GD, S includes both terms in (A.23).

B Empirical scope and methods

The figures answer different questions. Figure 4 establishes the training phenomenon. Figure 5 identifies the surviving force and its reinforcement on a matched continuation. Figures S2 and S3 test the conditional mechanism across targets. Their different training schedules and horizons are not pooled into a single theorem interval.

B.1 FULL-TRAINING OBSERVATION AND THE EARLY TRACKING TRANSITION

The target in Figure 4 is

$$f(x) = \frac{\sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(14\pi x)}{\sqrt{21/32}}, \quad x \in [-1, 1]. \quad (\text{B.1})$$

Width 512 corresponds to 467 reference interior cells and 22 halo centers per side, with $h = 2/467$. Training uses 2,048 uniform midpoint inputs, 4,096 validation points, and 8,192 dense evaluation points. The dense grid checks resolution; it is not an untouched generalization test. Training is deterministic, full batch, and FP64. Joint runs train all parameters from five paired random initializations for five million updates. One recipe per optimizer is selected by median final validation error across the five seeds. The selected full-horizon cosine schedules start at rates 0.05 for Adam and 0.2 for GD. No first-tolerance crossing replaces the endpoint rule.

Fixed GD trains a zero-initialized readout on supplied uniform $\lambda = 1/4$ features, using the spectral step $1/(2\mu_1)$, where μ_1 is the largest readout-kernel eigenvalue. Fixed Adam uses the same features and zero initialization, with a final-validation-selected cosine schedule starting at 10^{-4} . Both execute five million updates. Budgets and error definitions match; learning rates and search spaces need not.

Figure 4 plots relative training error with attached readouts. Joint lines summarize medians over seeds and display bins; bands preserve seed ranges and recorded within-bin extrema, including spikes. Fixed-Adam shading is its within-bin range, not seed uncertainty. Fixed GD is subsampled from its executed trace. Joint Adam/GD dense-grid endpoint medians are 0.0018148 and 0.24393; fixed Adam’s is 6.62×10^{-7} , and fixed GD’s training error is 3.88×10^{-4} . Joint Adam/GD RMS relative slopes are 0.05818 and 0.007177. Adam passes 1% relative error, so the claim is its remaining precision gap, not failure at that tolerance. The scale reference is not a theorem that all accurate representations must share that geometry.

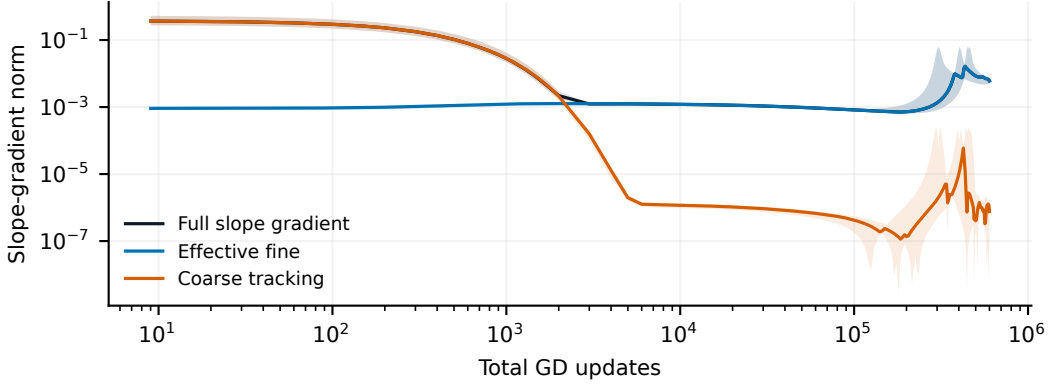


Figure S1: **Tracking loses its early dominance under GD.** A separate mixed-sine experiment at width 177, five seeds, and learning rate 0.002 over 600k updates. Curves show median slope-block norms; shading shows seed ranges. This full-history illustration does not determine the restart time for Figure 5. Late effective-force growth remains possible.

Figure S1 uses 2,048 training inputs and the same sine mixture normalized by training-grid RMS. Saved pre-update coordinates end at 599,999. Every plotted coarse solve is resolved; the numerical unresolved channel is zero. The curves are slope-block norms, whereas Figure 5 uses full-parameter norms to match the force identity. A crossing of these curves is not by itself a bound on accumulated tracking effects or a universal restart rule.

B.2 MATCHED POST-TRANSIENT CONTINUATIONS

The six-target audit uses width $W = 705$, reference resolution 512, $h = 2/512$, seed 30, and 2,048 uniformly spaced midpoint inputs. Initial slopes, hidden biases, and readouts are independent uniform draws on $[-\sqrt{6/(W+1)}, \sqrt{6/(W+1)}]$, with output bias zero; initialization is paired across targets. All parameters train with full-batch GD at rate 0.002. Restarts are the unmodified states after 20k updates, followed for another 100k updates. No readout refitting or final-accuracy checkpoint selection is used. The reference spacing normalizes scale; it does not describe an actual grid of the learned centers.

The targets before training-grid RMS normalization are

$$\begin{aligned}
 \text{Degree five: } & 0.3q_0 + 0.4q_1 + \sqrt{0.75}q_5, \\
 \text{Mixed sine: } & \sin(2\pi x) + 0.5\sin(6\pi x) + 0.25\sin(14\pi x), \\
 \text{Gaussian: } & \exp(-(x + 0.35)/0.22)^2, \\
 \text{Compact bump: } & \begin{cases} \exp(1 - 1/(1 - u^2)), & |u| < 1, \\ 0, & |u| \geq 1, \end{cases} \quad u = (x - 0.35)/0.22, \\
 \text{Smooth step: } & \tanh(14(x - 0.31)), \\
 \text{Kink: } & |x + 0.23|.
 \end{aligned}$$

Here q_k are empirical orthonormal polynomials with positive leading coefficient. The polynomial target is already normalized and is not x^5 . The other targets are divided by their original training-grid RMS, held fixed thereafter. The kink and compact bump test the training mechanism beyond the analytic target class used for the numerical construction; the persistence theorem itself only requires $f \in L^2(\mu)$.

Effective flow uses RK4 steps 0.02 and 0.01. GD uses rates 0.002 and 0.001 over the same elapsed flow time $T = 200$; the smaller-step control therefore uses 200k updates. There are 201 saved scalar diagnostic times. These controls test discretization sensitivity at fixed flow time; they do not compare equal optimizer update counts.

Both panels of Figure 5 use the smooth-step continuation. GD tracking norm is at most 0.1963% of effective-force norm at saved times. The maximum relative discrepancy between GD and paired effective-flow force norms is 0.0795%. Trapezoidal integration of the effective-flow rates reconstructs the log-force change with defect below 3.16×10^{-6} . Integrated feedback is 0.79164 and relaxation contributes -0.009283 : force grows from 0.001628 by a factor 2.187, while final relative error is 0.4897. The horizontal axes give total training age, adding the 20k restart age to elapsed flow time divided by 0.002. This example demonstrates limited amplification with positive reinforcement; it does not assert contraction of each slope.

B.3 CHECKING THE FEEDBACK CONDITION AND ITS USEFULNESS

Every target uses $B_t = 4d_{\text{dir}}(0)t$ in Figures S2 and S3. Four is drawn from the tested allowance family 1, 2, 4, 8; coverage is an empirical finding, not an initial-data guarantee. The largest sampled ratio of accumulated feedback to allowance is 0.54351.

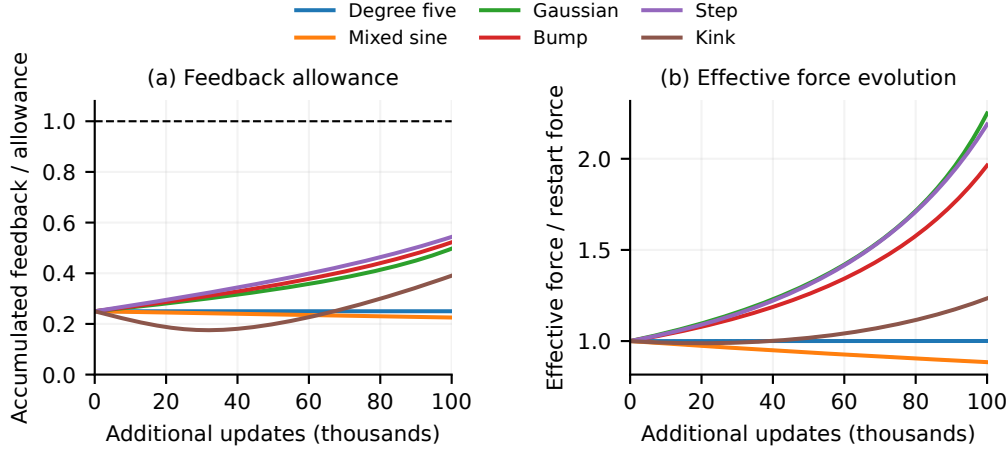


Figure S2: **The feedback condition permits either force decay or growth.** Six targets at width 705, from the 20k restart through 100k additional GD updates. Left: accumulated directional feedback divided by its common allowance $4d_{\text{dir}}(0)t$. Right: normalized force along GD (solid) and effective flow (dotted). Their agreement checks the reduction; it is distinct from checking the feedback allowance.

The effective-flow relative-error floors range from 0.39288 to 0.9124. RMS slope-displacement allowances range from 2.51×10^{-7} to 7.74×10^{-4} , compared with observed GD displacements from 2.61×10^{-8} to 5.25×10^{-5} . For increment 0.125, the largest conditional ever-acquired-fraction bound is 3.84×10^{-5} . All starting relative slopes are below 0.001; this increment is already less than needed to reach the supplied benchmark 0.25. The independent error floor is essential because that benchmark is not a necessary criterion for accuracy. The lenient 1% reference in this six-target audit is distinct from the long Adam experiment, which passes it.

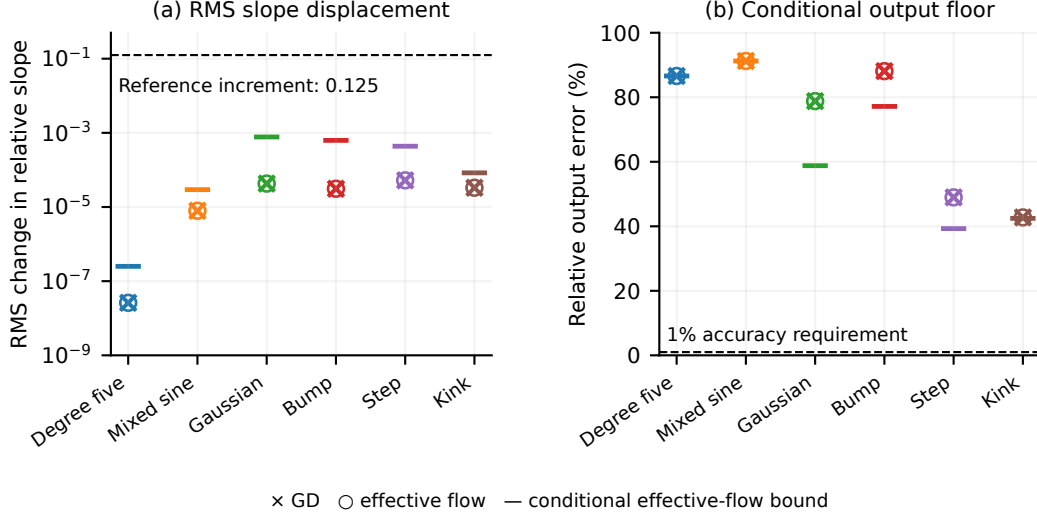


Figure S3: **The condition gives useful population and output bounds.** The same six continuations as Figure S2. Left: endpoint RMS displacement of relative slopes, with its effective-flow upper bound; this is displacement from the restart, not the RMS level in Figure 4. Right: relative output errors with attached readouts and effective-flow lower bounds. These rounded numerical evaluations set $S = 0$; GD agreement is measured separately. None is an interval certificate.

The broader width-705 baseline contains 23 targets and two seeds, spanning polynomial, oscillatory, localized, rational, exponential, and transition functions. Twice the restart directional rate covers every saved prefix of all 46 branches over 20k further updates; the largest required multiplier is 1.367. Twelve width-1409 reference branches also pass. These include related target variants; the six dense continuations are part of this investigation, not six extra independent families.

Across the six targets, the largest refined GD/effective-flow endpoint relative-error difference is 1.72×10^{-6} . Sampled accumulated tracking-induced derivative loading is at most 0.0993% of restart force. Including sampled tracking effects in a sharper time-integrated energy calculation changes its relative-error floor by at most 4.16×10^{-6} ; this is not a certified value of Δ_{err} in (A.21). Halving the GD step changes endpoint parameters by less than 1.98×10^{-6} in Euclidean norm; halving the RK4 step changes them by less than 2.4×10^{-14} .

The broad archive is sparse; the matched continuations are sampled densely. Neither provides a rigorous enclosure between every saved state or for the finite-step defect in (A.23). The conditional theorem is exact, whereas verification of its premises is numerical. Force and error use the training measure and actual attached readouts; no continuous-input generalization or quantitative Adam guarantee is claimed.

B.4 EMPIRICAL COVERAGE OF THE POPULATION FEEDBACK LOOP

The stronger theorem uses $b_0 = d_{\text{dir}}(0)$ and $C(t) = 2\sqrt{I_F(0)}t$. At saved prefixes, the smallest required response coefficient is

$$K_{\text{req}} = \max_{t_i > 0} \frac{[\mathcal{B}(t_i) - b_0 t_i]_+}{\int_0^{t_i} \mathcal{A}_4(u) du}. \quad (\text{B.2})$$

We compare it with $K_{\text{crit}} = 1/(es_0 G(T))$. This is a retrospective check that useful constants exist, rather than a prediction from the restart alone. Integrals use scalar interpolants between saved states. The accumulated concentration allowance passes on all reported paths.

All 46 width-705 paths across 23 targets pass the sampled loop criterion over 20k further updates; the largest $K_{\text{req}}/K_{\text{crit}}$ is 0.1065. All 12 width-1409 reference paths pass too. All six densely sampled

effective-flow continuations pass over that shorter duration. At 100k further update-equivalent units, degree five, mixed sine, and kink still pass. Gaussian, bump, and step give ratios 2.67, 3.12, and 2.24; the broader feedback-budget bounds remain informative on all six.

For these three longer failures, the required K rises only by factors 1.184, 1.131, and 1.002 compared with its first-20k estimate. The duration-dependent sufficient criterion can therefore expire while the response relation changes little. Doubling a coefficient fitted only on the first 20k further updates covers the longer response on five of six targets. Kink is the exception: its early excess feedback is zero, but later becomes positive. These results support finite-interval response conditions without supplying a universal extrapolation rule. Halving RK4 and GD steps preserves the pass/fail classifications. GD diagnostics test the premises numerically; they do not certify the disturbed criterion in (A.35).

C Interventions and the domain of the mechanism

Function-preserving fourfold cloning divides each readout among four identical neurons. It preserves output while dividing each copy’s geometry gradient by four and multiplying aggregate readout mobility by four. Correcting both learning rates recovers the original trajectory to numerical precision. Across the tested 23-target cohorts, correcting geometry mobility largely restores slope motion, whereas correcting readout mobility alone does not. Faster readout fitting alone is insufficient to explain this controlled slowdown.

The smallest tested outward pulses retain 99.0–100.3% of their directional offset after 20k updates. This supports slow motion without strong restoration in those tested directions, not a global attractor. Large geometry injections can leave the weak-force regime: under the primary coarse-repair procedure, median initial effective-force norm rises from about 6.04×10^{-4} at baseline to 2.64 at 100-fold injection. The corresponding weak-force error bound becomes uninformative; that alone does not show successful training.

These interventions constrain the explanation without imposing one signed force contribution across targets. Correcting generated lower-order errors can oppose expansion, but neither universal slope contraction nor that specific error pattern is an assumption of Theorem A.1. The claim is limited useful reinforcement over a stated interval, while the network remains free to evolve.